Algorithm Analysis: LPCA and CDC

Formulations, Similarities, and Structural Differences

Local Principal Component Analysis (LPCA)

$$\mathcal{X} = \bigcup_{k=1}^{K} \mathcal{X}_k, \quad \mathcal{X}_k \cap \mathcal{X}_j = \emptyset$$

$$\boldsymbol{\mu}_k = \frac{1}{|\mathcal{X}_k|} \sum_{\boldsymbol{x} \in \mathcal{X}_k} \boldsymbol{x}$$

$$\boldsymbol{\Sigma}_k = \frac{1}{|\mathcal{X}_k|} \sum_{\boldsymbol{x} \in \mathcal{X}_k} (\boldsymbol{x} - \boldsymbol{\mu}_k)(\boldsymbol{x} - \boldsymbol{\mu}_k)^T$$

$$\boldsymbol{\Sigma}_k \boldsymbol{U}_k = \boldsymbol{U}_k \boldsymbol{\Lambda}_k \quad (\text{where } \boldsymbol{U}_k \in \mathbb{R}^{d \times q})$$

$$\hat{\boldsymbol{x}}_i = \boldsymbol{\mu}_k + \boldsymbol{U}_k \boldsymbol{U}_k^T (\boldsymbol{x}_i - \boldsymbol{\mu}_k)$$

$$\min_{\boldsymbol{\mu}, \boldsymbol{U}} L_{LPCA} = \sum_{k=1}^{K} \sum_{\boldsymbol{x}_i \in \mathcal{X}_k} \left\| (\boldsymbol{I} - \boldsymbol{U}_k \boldsymbol{U}_k^T)(\boldsymbol{x}_i - \boldsymbol{\mu}_k) \right\|_2^2$$

Carré du Champ (CDC) Operator

Markov Triple: $(\mathcal{M}, \mu, \Gamma)$, Infinitesimal Generator: L

$$\Gamma(f, h) = \frac{1}{2}\big(fLh + hLf - L(fh)\big)$$

$$g(df, dh) = \Gamma(f, h) \quad \text{(Riemannian metric recovery)}$$

Discrete Approximation (Markov Chain Generator L):

$$\Gamma(f, h)(x_i) = -\sum_{j=1}^{n} L(i, j)\big(f(x_i) - f(x_j)\big)\big(h(x_i) - h(x_j)\big)$$

$$L\phi_i = \lambda_i \phi_i \quad \text{(Eigendecomposition of Generator)}$$

Mathematical Formulations: LPCA vs. CDC

Local PCA (LPCA)

Local Covariance for region $\mathcal{X}_k$:

$$\boldsymbol{\Sigma}_k = \sum_{\boldsymbol{x}_j \in \mathcal{X}_k} \frac{1}{|\mathcal{X}_k|}(\boldsymbol{x}_j - \boldsymbol{\mu}_k)(\boldsymbol{x}_j - \boldsymbol{\mu}_k)^T$$

Anchor: $\boldsymbol{\mu}_k$ (Local Mean)

Weights: $\frac{1}{|\mathcal{X}_k|}$ (Uniform)

Carré du Champ (CDC)

Metric at specific point $\boldsymbol{x}_i$:

$$\boldsymbol{\Gamma}(\boldsymbol{x}_i) = \sum_{j=1}^{n} P_{ij}(\boldsymbol{x}_i - \boldsymbol{x}_j)(\boldsymbol{x}_i - \boldsymbol{x}_j)^T$$

Anchor: $\boldsymbol{x}_i$ (The Point Itself)

Weights: P_{ij} (Markov Transitions)

Similarities: LPCA and CDC

- **Algebraic Structure:** Both construct a positive semi-definite matrix using the outer product of displacement vectors.
- **Spectral Analysis:** Both recover the local tangent space and intrinsic dimensionality via the eigendecomposition of their respective matrices.
- **Dimensionality:** Both operate in the ambient dimension $\mathbb{R}^{D \times D}$.

Structural Differences

Feature	Local PCA	Carré du Champ (CDC)
Matrix Type	Regional Covariance $\boldsymbol{\Sigma}_k$	Pointwise Metric Tensor $\boldsymbol{\Gamma}(\boldsymbol{x}_i)$
Displacement	Point to Mean $(\boldsymbol{x}_j - \boldsymbol{\mu}_k)$	Point to Neighbours $(\boldsymbol{x}_i - \boldsymbol{x}_j)$
Neighbourhood	Hard partitioning $(\boldsymbol{x}_j \in \mathcal{X}_k)$	Soft probabilistic bounds (P_{ij})
Geometry	Assumes flat local patches	Recovers curved Riemannian manifold
Robustness	Highly sensitive to outliers	Robust via diffusion kernel